

Water security and sustainable management of the Kura River for future generations

Aydan Muradzade¹, Nuray Hasanova², Tunzale Mammadova^{3*}

¹Secondary School No. 1, Sumgait, Azerbaijan

²I. Gayibov Secondary School No. 1, Sumgait, Azerbaijan

³The Modern Educational Complex, Baku, Azerbaijan

*Corresponding author: tunzale.mammadova@mtk.edu.az

Article Information

Received: August 5, 2026

Revised: August 31, 2026

Accepted: September 10, 2026

Published: September 20, 2026

Academic Editor: Prof. Dr. Elnur Sadiqov

Copyright: © 2026 by the authors.

This article is distributed under the terms of the [Creative Commons](https://creativecommons.org/licenses/by/4.0/)

[Attribution 4.0 International License](https://creativecommons.org/licenses/by/4.0/) (CC BY 4.0).

Keywords

Kura River, sustainable water management, water conservation, water security, water-use efficiency

ABSTRACT

The Kura River is central to Azerbaijan's water security because it supports drinking-water supply, irrigation, energy production and riverine ecosystems while remaining strongly dependent on transboundary inflows. This study examines how water scarcity, inefficient use and environmental pressure are understood at the local research level and how these concerns align with contemporary water-management priorities. The paper integrates two independently prepared 2026 research dossiers with a targeted review of official and peer-reviewed literature. One source project used desk research, maps and a five-factor problem prioritization, while the second combined desk research with an exploratory questionnaire and interview material. The source-based prioritization assigned the greatest emphasis to pollution (35%), followed by water overuse (30%), climate change (20%), solid waste (10%) and riparian tree loss (5%). In the exploratory survey material, 63% identified water waste as the leading cause of depletion, 80% reported avoiding unnecessary water use, 63% selected drinking-water shortage as the principal anticipated consequence, and 49% preferred stronger oversight of water consumption as the most effective measure. Because the source files do not document the survey sample size or respondent-level data, these percentages are interpreted descriptively rather than as population estimates. The combined evidence supports an integrated strategy based on infrastructure modernization, efficient irrigation, wastewater control, rainwater and treated-water reuse, public awareness, monitoring and basin-scale cooperation. These priorities are broadly consistent with Azerbaijan's 2024–2040 National Strategy on efficient use of water resources.

INTRODUCTION

Water security is not limited to the physical availability of water. It also depends on whether sufficient water of acceptable quality can be supplied to households, agriculture, ecosystems and the economy while drought, pollution and other water-related risks remain manageable (Grey & Sadoff, 2007). This broader perspective is particularly relevant to Azerbaijan, where the Kura River is the principal surface-water artery and where water availability is shaped by both domestic demand and upstream conditions. The World Bank (2024) estimates that roughly 70% of Azerbaijan's surface-water flows originate outside the country, making national water security highly sensitive to transboundary hydrology, upstream abstraction and basin-wide environmental pressures.

The Kura River therefore connects several dimensions of sustainability at the same time. Its water is used for municipal supply, irrigated agriculture, hydropower and ecosystem maintenance, while the river and its tributaries are exposed to pollution, seasonal variability, infrastructure losses and increasing demand. The World Bank (2024) identifies irrigated agriculture as the largest water-consuming sector in Azerbaijan and notes that inefficient irrigation, aging infrastructure, pollution and drought risk remain important constraints. FAO (2021) similarly emphasizes that declining rainfall, higher temperatures, reduced runoff and old irrigation systems have intensified water-scarcity concerns, particularly

because a large share of the country's usable water originates beyond its borders.

The two original research works integrated in this paper approached these issues from a future-oriented perspective. Muradzade's project focused on protecting water resources for future generations and organized the main Kura-related pressures into pollution, excessive water use, climate change, solid waste and riparian vegetation loss. Hasanova's project examined water-management strategy, irrigation losses, rainwater harvesting, alternative water use and public attitudes toward conservation. Both projects emphasized that household behavior and infrastructure management are linked: individual conservation can reduce avoidable demand, but long-term water security also requires institutional planning, irrigation modernization, water-quality protection and climate adaptation.

The objective of this study is to convert those source-level observations into a structured academic assessment of Kura River water security. Three questions guide the analysis: Which pressures are most prominent in the original research materials? What do the exploratory perception data suggest about public understanding and water-saving behavior? How do the proposed solutions correspond to current evidence and Azerbaijan's national water-management strategy? The study does not reinterpret the source percentages as nationally representative statistics; instead, it uses them as descriptive

evidence and evaluates them against recent scientific and institutional literature.

LITERATURE REVIEW

Water security is increasingly treated as a coupled environmental, economic and governance problem. Grey and Sadoff (2007) define the concept around adequate quantity and quality of water for health, livelihoods, ecosystems and production together with acceptable water-related risks. This framing is useful for the Kura because scarcity cannot be separated from water quality, irrigation efficiency, drought exposure or institutional coordination. In the Kura-Aras context, Suleymanov (2025) argues that stronger basin institutions and cooperative governance are important for managing shared water resources, particularly where scarcity and transboundary dependence interact.

Recent assessments of Azerbaijan reinforce this integrated view. The World Bank's General Water Security Assessment identifies transboundary dependence, seasonal variability, inefficient irrigation, infrastructure condition and pollution as interrelated challenges and recommends stronger resource management, modernization and adoption of efficient technologies (World Bank, 2024). FAO (2021) likewise highlights old earthen canals and associated distribution losses as important obstacles to sustainable agricultural water use. Earlier research by Rzaev (2017) also concluded that irrigation-system modernization and better agricultural water management are necessary to prevent depletion under growing domestic demand and reduced transboundary inflows.

Water quantity and water quality are also connected. Aliyev and Abduev (2025), using monitoring data from 2020–2022, documented substantial downstream changes in the Kura River's mineralization and major-ion composition. Their findings show why sustainable management cannot focus only on the volume of water stored or diverted: lower flows, wastewater inputs, agricultural drainage and tributary chemistry can alter the quality of water available for drinking, irrigation and ecosystems. Protecting the Kura therefore requires both demand-side efficiency and water-quality control.

A major institutional development is Azerbaijan's National Strategy on Efficient Use of Water Resources for 2024–2040, approved in October 2024. The strategy is organized in three stages and addresses integrated water-resource management, reliable drinking and irrigation supply, infrastructure modernization, reuse of treated water, rainwater management, drought and flood risks, and hydroecological protection (President of the Republic of Azerbaijan, 2024). Several elements correspond directly to proposals in the source projects, including rainwater collection, modernization of irrigation systems and stronger conservation awareness.

Taken together, the literature suggests that future-oriented Kura management should combine four functions: protecting the resource base, reducing avoidable demand, improving infrastructure and water-use efficiency, and strengthening governance and monitoring. The present study uses these functions as the analytical framework for organizing the original research materials and distinguishing source-derived observations from broader evidence in the literature.

MATERIALS AND METHODS

The study used an integrative qualitative design based on two independently prepared 2026 research dossiers. The first dossier, prepared by Aydan Muradzade, examined protection of water resources for future generations through online research, map and image review, and a source-level prioritization of five pressures

affecting the Kura River. The second dossier, prepared by Nuray Hasanova, focused on water-management strategy and combined desk research with questionnaire and interview material, including charts on perceived causes of water depletion, personal conservation practices, expected consequences and preferred management measures.

The source materials were coded into four analytical domains: (1) water availability and climate exposure; (2) demand, irrigation and infrastructure efficiency; (3) pollution and ecosystem protection; and (4) governance, public behavior and adaptation. Muradzade's five-factor percentages were retained exactly as reported but are treated as a project-level prioritization rather than as survey estimates because the source document does not describe a statistical procedure for generating them. Hasanova's questionnaire percentages were transcribed from the original charts. The source file does not report the survey sample size, and it contains a textual reference to interviews with children aged 5–11 while separate charts display a broader age distribution. For that reason, the questionnaire results are used only as exploratory descriptive evidence and no inferential statistics are applied.

A targeted literature review was used for triangulation. Priority was given to Azerbaijan's 2024–2040 National Strategy, the World Bank's recent national water-security assessment, FAO material on agricultural water governance, and peer-reviewed studies addressing Kura-Aras governance, irrigation and water quality. The comparison assessed whether source-level concerns and proposed solutions were consistent with documented national and basin-level challenges. No new field sampling, laboratory analysis or respondent-level survey processing was undertaken for this article, and no personal identifiers were used.

RESULTS AND DISCUSSION

Water-security pressures identified in the source research

Muradzade's source project assigned the highest relative emphasis to pollution (35%) and water overuse (30%), followed by climate change (20%), solid waste (10%) and loss of riparian trees (5%). These percentages should not be interpreted as measured contributions to Kura River decline; rather, they provide a transparent record of how the original research prioritized the problem. The ranking is nevertheless consistent with the broader literature in one important respect: Azerbaijan's water-security challenge is multi-causal. The World Bank (2024) identifies abstraction, irrigation inefficiency, pollution and climate-related risk simultaneously, while the National Strategy treats infrastructure, water quality, drought and efficient use as linked management areas.

Pollution was the most strongly emphasized pressure in Muradzade's work. The project linked untreated wastewater and litter to reduced suitability of river water and ecological damage. Recent monitoring confirms that water-quality conditions vary substantially along the Kura. Aliyev and Abduev (2025) reported a strong downstream mineralization gradient and changes in major-ion composition, demonstrating that the river cannot be treated as chemically uniform from the upstream border to the lower reaches. For management, this means that water security requires location-specific water-quality monitoring in addition to overall allocation planning.

The second major pressure was inefficient or excessive water use. Both source projects repeatedly discussed open-channel losses, outdated irrigation practices and avoidable household consumption. These concerns align closely with FAO (2021), which notes significant losses in older irrigation infrastructure, and with Rzaev (2017), who identifies irrigation modernization as central to water-resource

protection in Azerbaijan's arid zones. Modernization should therefore be understood not only as an agricultural productivity measure but also as a water-security intervention that can reduce the volume withdrawn from river systems for the same productive purpose.

Climate change was the third major source theme. The student projects referred to hotter summers, lower rainfall, reduced snow availability and greater risk of future scarcity. The National Strategy itself cites projected reductions in national water resources under warming scenarios and places drought management and climate-resilient water supply among its long-term priorities (President of the Republic of Azerbaijan, 2024). Climate pressure can also amplify other problems: when flows are reduced, competing demands become more difficult to reconcile and water-quality pressures may become more consequential because dilution capacity is lower.

Source-derived prioritization and exploratory perception evidence

Table 1 preserves Muradzade's original five-factor weighting while translating each factor into a management implication. The purpose is not to quantify causal responsibility but to show how the source research connected environmental problems with proposed action. Pollution and overuse together account for 65% of the project-level emphasis, which explains why the original recommendations concentrated on wastewater control, household conservation, efficient irrigation and environmental education.

Hasanova's questionnaire material provides a complementary behavioral perspective. Among the reported responses, 63% selected water waste as the main cause of depletion, 80% stated that they avoid unnecessary water use, 63% identified drinking-water shortage as the main anticipated consequence, and 49% selected stronger oversight of water consumption as the preferred management measure. Other responses in the source charts included support for awareness activities, new technologies and penalties. Because the number and composition of respondents are not fully documented, these values should be interpreted as indicators of the concerns present in the project sample, not as estimates for Sumgait or Azerbaijan.

The perception data are still useful for understanding the logic of the original research. Respondents appear to connect water security primarily with everyday waste and future drinking-water availability, while the preferred responses combine behavioral control and public education. This is important because infrastructure investment alone cannot eliminate inefficient use if households and water users do not understand scarcity. Conversely, awareness campaigns alone cannot compensate for leakage, inefficient irrigation or untreated wastewater. Sustainable management therefore requires a combination of demand-side behavior and system-level modernization.

Table 1. Source-project problem weighting and corresponding management relevance

Pressure	Reported emphasis and management relevance
Pollution	35% — Wastewater treatment, source control and water-quality monitoring
Water overuse	30% — Demand management, leakage reduction and irrigation efficiency
Climate change	20% — Drought preparedness, climate-resilient allocation and storage
Solid waste	10% — Waste prevention, riverbank management and public awareness
Riparian tree loss	5% — Riparian restoration, erosion control and ecosystem protection

Alignment with Azerbaijan's 2024–2040 water strategy

The policy alignment between the student research and the National Strategy is notable. The strategy's first stage (2024–2027) emphasizes

institutional and infrastructure improvements and includes actions on wastewater and drainage-water reuse, rainwater management, irrigation-system reconstruction, alternative drinking-water sources, drought measures and pollution-risk reduction. Its later stages extend the framework toward 2040. The source projects proposed many of the same practical directions in simpler form: rainwater harvesting, drip irrigation, reduced household waste, wastewater treatment, tree planting and stronger environmental education. This convergence suggests that the students identified management themes that are relevant beyond the school-project context.

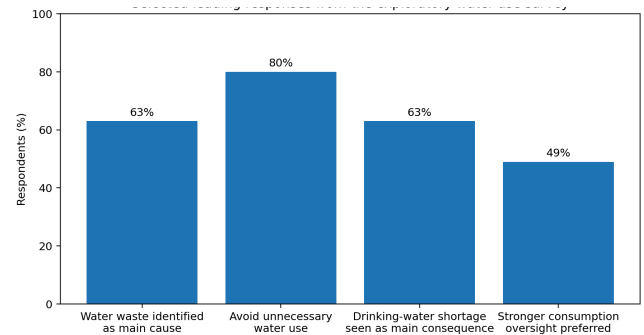


Figure 1. Selected leading responses from the exploratory water-use survey reported in the source research

Figure 1 summarizes four of the clearest majority responses recorded in Hasanova's charts. The dominant behavioral message is that respondents recognize avoidable water use as a problem and report some personal conservation behavior, yet they also expect future drinking-water shortages and favor stronger oversight. The apparent coexistence of awareness and concern implies that knowledge does not automatically resolve water insecurity. Effective management must create practical conditions in which conservation is easy, measurable and supported by infrastructure, pricing and service reliability. The figure is descriptive only because the source material does not provide the sample size or respondent-level observations.

Integrated management priorities for future generations

Five priorities emerge from the combined analysis. First, irrigation and distribution infrastructure should be modernized to reduce conveyance and application losses, supported by measurement and efficient technologies such as drip irrigation where technically and economically appropriate. Second, wastewater treatment and water-quality monitoring should protect the usable resource, especially in downstream reaches. Third, alternative and circular water sources should be expanded where feasible, including rainwater harvesting and safe reuse of treated wastewater for non-potable applications; both are explicitly reflected in the National Strategy. Fourth, drought planning and climate adaptation should connect hydrological monitoring with flexible allocation rules so that scarce water can be managed transparently during low-flow periods. Fifth, public awareness should be linked to measurable conservation practice in schools, households and agricultural communities. Because the Kura is transboundary, these domestic actions should be complemented by basin-level information exchange and cooperation. Together, these measures frame water conservation not as a single campaign but as a long-term system for protecting quantity, quality and access for future generations.

Limitations and future research priorities

The main limitation of this study is inherited from the original research materials. The first source project provides a transparent problem-weighting scheme but does not describe how the percentages were calculated; they therefore represent the author's structured

prioritization rather than measured shares of hydrological or environmental causation. The second source project contains useful questionnaire charts, yet the sample size, recruitment procedure and respondent-level dataset are not documented. The textual description also refers to interviews with children aged 5–11, whereas a separate chart shows respondents from several age groups. These inconsistencies do not invalidate the educational value of the project, but they prevent statistical generalization. A future version of the research should record the exact number of participants, sampling location, age group, question wording and response counts and should preserve an anonymized dataset so that percentages can be independently verified. If minors participate, consent and school-level ethical procedures should also be documented explicitly.

A second limitation is the absence of direct hydrological and water-quality measurements in the two source dossiers. Perceptions of scarcity and observations of pollution should ideally be tested against measurable indicators such as river discharge, seasonal water level, electrical conductivity, salinity or mineralization, selected nutrients and basic water-quality parameters. Repeated observations at fixed locations would make it possible to distinguish seasonal variation from longer-term change. For a school-based monitoring programme, this need not begin with advanced laboratory analysis: standardized field sheets, georeferenced photographs, simple water-level observations and carefully selected low-cost sensors can create a reproducible baseline when quality-control procedures are followed. These observations could then be compared with official monitoring and published studies such as Aliyev and Abduev (2025). A third priority is to connect perception research with actual water-use behavior. Household or school water-meter readings before and after an awareness intervention, for example, would allow researchers to test whether knowledge produces measurable conservation. Such a design would preserve the practical and youth-oriented character of the original projects while providing stronger evidence for future water-security planning.

CONCLUSIONS

This study integrated two original research projects into an academic assessment of Kura River water security and sustainable management. The source materials consistently identified pollution, inefficient water use, climate change, solid waste and degradation of riparian vegetation as threats to the future of the river. Muradzade's project-level weighting placed pollution and water overuse at the center of the problem, while Hasanova's exploratory questionnaire showed strong concern about water waste and future drinking-water shortage and substantial reported engagement in basic conservation behavior. These percentages are useful as descriptive project evidence but cannot be generalized because the source files do not document a survey sample size or provide respondent-level data. The wider literature nevertheless confirms the relevance of the themes identified by the students. Azerbaijan remains highly dependent on transboundary surface water, irrigated agriculture is a major user, old infrastructure can generate substantial losses, and both water quantity and quality are affected by climate, pollution and management conditions. The 2024–2040 National Strategy provides a policy framework that closely overlaps with the source recommendations,

including irrigation modernization, rainwater management, wastewater reuse, drought management and stronger resource governance. The central implication is that protecting the Kura for future generations requires integrated action across infrastructure, water quality, efficient demand, climate adaptation, education and transboundary cooperation. Future work should improve the empirical basis of the school research by documenting sample design, collecting georeferenced observations and linking perception surveys with measured indicators of water use and river condition.

AUTHOR CONTRIBUTIONS

A.M. and N.H. contributed the original research materials, problem identification, source collection and preliminary analysis. N.H. also contributed the exploratory questionnaire and interview materials. T.M. provided supervision, methodological guidance, academic integration and review and editing. All authors contributed to interpretation, reviewed the final manuscript and approved the submitted version.

CONFLICTS OF INTEREST

The authors declare no conflict of interest.

FUNDING

This research received no external funding.

DATA AVAILABILITY

The source-level charts, percentages and qualitative research materials used in this synthesis are described in the article. Respondent-level survey data and a documented sample size were not available in the source files; therefore, no inferential statistical dataset is provided.

REFERENCES

- Aliyev, E., & Abduev, M. (2025). Quality trend of Kura River in Azerbaijan by major ion concentration. *Discover Water*, 5, 82. <https://doi.org/10.1007/s43832-025-00219-9>
- Food and Agriculture Organization of the United Nations. (2021). Improved water governance: Towards sustainable agricultural development, Azerbaijan. FAO.
- Grey, D., & Sadoff, C. W. (2007). Sink or swim? Water security for growth and development. *Water Policy*, 9(6), 545–571. <https://doi.org/10.2166/wp.2007.021>
- President of the Republic of Azerbaijan. (2024). National Strategy on Efficient Use of Water Resources, 2024–2040. Approved by Presidential Order No. 320 of 10 October 2024. <https://president.az/az/articles/view/67069>
- Rzayev, M. A. (2017). Problems of water protection against adverse impact of irrigated agriculture and approaches to their solution in arid zones. *Water Resources*, 44(1), 158–166. <https://doi.org/10.1134/S0097807817010134>
- Suleymanov, F. (2025). The institutionalization of the Kura-Aras River Basin for effective management of water resources. *International Journal of River Basin Management*, 23(4), 675–685. <https://doi.org/10.1080/15715124.2024.2363310>
- World Bank. (2024). Azerbaijan: General water security assessment. World Bank Group.
- World Bank. (2024). Azerbaijan - Transitioning to efficient water sector institutions and programs for addressing water security challenges. World Bank Group.
- Food and Agriculture Organization of the United Nations. (2008). AQUASTAT country profile: Azerbaijan. FAO.
- Azerbaijan National Encyclopedia. (2007). Kura River. In *Azerbaijan National Encyclopedia* (Vol. 14). Baku.